

Module E (WIP): a neutral null for ancestry sorting, and what the empirical population structure already rules out

Work in progress — July 29, 2026

Abstract

This is a working note on Module E: a recombination-matched, haplodiploid *neutral null* for the 20 replicate *F. polycтена* $\times$ *F. aquilonia* hybrid populations, built to test whether the directional ancestry sorting seen in Modules A–C exceeds drift. The null conditions on real phased parental haplotypes, uses the empirical recombination map, is haplodiploid, and contains no selection. The central design choice is to stratify markers by DiagnosticIndex (DI): low-DI markers cannot be driven by ancestry sorting and so serve as an *internal demographic control* in the same analysis that tests the high-DI markers. Two results now stand largely on the empirical data alone, independent of the simulation calibration. First, among-population differentiation scales steeply with DI in the data (F_{ST} : ~ 0 at neutral markers $\rightarrow 0.31$ at diagnostic markers) whereas neutral drift, being DI-blind, is flat. Second, decomposing the diagnostic differentiation shows that a single per-population admixture proportion (“variable founding admixture”) explains only $\sim 6\%$: the differentiation is 78% *locus-specific*. Because neutral drift is excluded (neutral $F_{ST} \approx 0$) and variable admixture is excluded (6%), what remains is a large, locus-specific, *divergent* ancestry signal standing against a near-zero neutral expectation. A wrinkle worth flagging: this signal is divergent (populations differing at diagnostic loci), which is distinct from — though compatible with — the parallel directional sorting of Module A. The definitive test, a p -value from the sorting statistic run through the calibrated null, is in progress.

1 The null model

Rather than simulating the parental species’ history, we *condition* on it: founders are drawn from real phased parental haplotypes (15 diploid females per species, imputed and phased within species with Beagle 5.4; pool of 30 *aquilonia* haplotypes and 13 *polycтена* diploid females), the mutation rate is zero, and only the post-contact hybrid phase is simulated. This inherits true allele frequencies, true between-species divergence, and true within-species haplotype structure. The engine is SLiM 5, non-Wright–Fisher, haplodiploid (haploid males from unfertilised eggs, spermathecal storage, Pamilo mating, age-structured life table), over the 26 assembled chromosomes with their empirical cM maps (converted with `scale`= 10^{-8}). All selection is removed, so drift and recombination on the empirical map are the only forces. Each parameter set runs as 20 independent demes sampled every 20 generations to generation 160.

Requirements for a valid comparison. Three points are load-bearing and worth recording. (i) **Sample-size matching:** composite r^2 and the sorting classification are both sample-size sensitive, so simulated demes are subsampled to the empirical per-population sizes (3–20 diploids). (ii) **Symmetric treatment:** any filter applied to the data is applied to the null (“sorted” *is* high differentiation, so filtering the data alone would move it toward the null mechanically). (iii) **F_{ST} estimator:** with per-population n of 3–20, the Nei estimator is upward-biased by $\mathcal{O}(1/2n)$; the Hudson estimator corrects this and is used for the structure results below. This correction matters (Section 3).

Neutral alternatives excluded as controls. Before interpreting any excess we quantified and excluded several neutral explanations. *Founder heterogeneity* (populations founded from different local sub-pools): even the absolute ceiling — each deme founded from a single *aquilonia* haplotype and a single *polycytena* female — gives an at-founding among-population F_{ST} of only 0.090, because at diagnostic markers both species are near-fixed for opposite alleles and a 50/50 admixture washes out within-species structure. *Colony (nest) sampling*: measured within-nest excess genotype correlation is only 0.047 (polygyny), so sibling structure is a minor effect. *Mixed population ages*: age mixtures interpolate along the LD-versus-drift trade-off and match no summary. These are recorded so they need not be re-litigated.

2 Differentiation and sorting scale with DI in the data, not under drift

Markers were binned by DI over the parent-MAF ≥ 0.15 universe (40,000 markers, 4,000 per bin across ten fixed DI bins). Figure 1 and Table 1 give the result: observed among-population F_{ST} rises from the neutral background to the most ancestry-informative markers, while the neutral null is essentially flat. The fraction of sorted loci shows the same pattern more sharply, from a ~ 1.5 -fold excess at low DI to >100 -fold at high DI. Because drift does not distinguish informative from uninformative markers, a DI-scaled discrepancy cannot be produced by demographic misspecification — the comparison is internally controlled because both datasets use the same markers, MAF gate, and frequency spectrum. (Table 1 reports Nei F_{ST} ; the absolute neutral value there, 0.05, is the small-sample bias corrected in Section 3, but the *scaling* is unaffected.)

Table 1: Observed vs simulated statistics by DI bin at the anchor-matched neutral cell. “sorted” is the fraction of loci classed as directionally sorted.

DI bin	F_{ST} obs	F_{ST} null	π obs	π null	sorted obs	sorted null
≤ -90	0.051	0.191	0.448	0.381	0.026	0.021
$(-75, -60]$	0.303	0.183	0.260	0.314	0.350	0.233
$(-40, -30]$	0.305	0.167	0.277	0.364	0.260	0.072
$(-25, -20]$	0.310	0.160	0.297	0.398	0.150	0.015
> -15	0.315	0.156	0.304	0.415	0.116	0.000

3 Population structure: neutral ≈ 0 , diagnostic strong

This is the pivotal result. At *strictly* neutral markers (DI ≤ -90 , 13,978 markers), the pairwise Hudson F_{ST} among the 20 populations is essentially **zero** (median 0.000, max 0.012), and the neutral PCA is a single overlapping cloud with no population structure (Fig. 2, left and middle). At diagnostic markers (DI > -25) the same populations are **strongly structured**, $F_{ST} = 0.31$, resolving into distinct clusters (Fig. 2, right).

Two consequences follow. First, the earlier “neutral $F_{ST} \approx 0.05$ ” was small-sample bias in the Nei estimator; the Hudson estimate is ≈ 0 . The same bias inflated the *simulated* Nei values equally, so the sim-versus-data gap in Table 1 survives, but the true neutral differentiation is ≈ 0 . Second, and more importantly, this dissociation cannot be drift: drift is DI-blind, so it cannot differentiate diagnostic markers while leaving neutral markers undifferentiated. The populations do not behave as 20 independently drifting demes at neutral loci (independent foundings accumulate $F_{ST} \approx t/2N_e > 0$); they look effectively panmictic / very-large- N_e / connected at neutral loci, while differing strongly in ancestry. A Mantel-style test confirms the two are dissociated: Spearman correlation between pairwise neutral F_{ST} and pairwise |ancestry difference| is only 0.08.

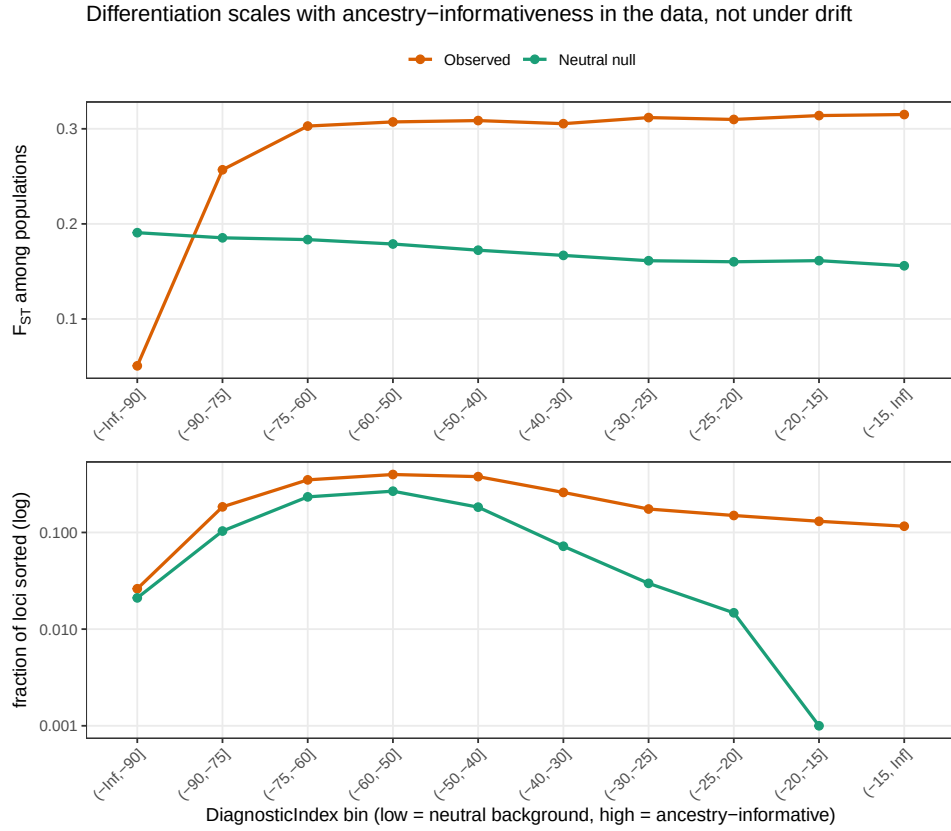


Figure 1: Among-population differentiation (top) and the fraction of sorted loci (bottom, log scale) against DI bin, observed vs neutral null. Neutral drift is DI-blind and produces an essentially flat profile; the data concentrate both differentiation and sorting in ancestry-informative markers.

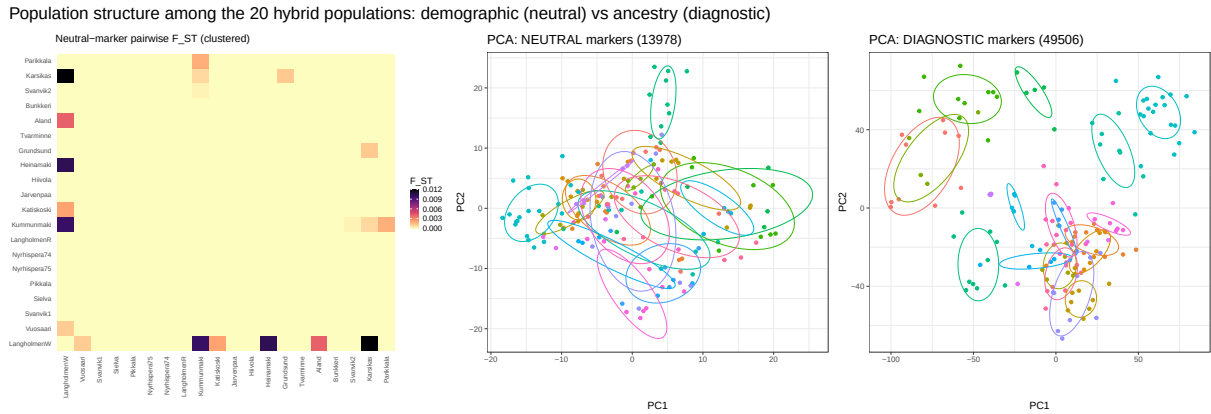


Figure 2: Structure among the 20 hybrid populations. Left: pairwise neutral-marker ($DI \leq -90$) F_{ST} , Hudson estimator, clustered — essentially zero everywhere. Middle: individual PCA on neutral markers — one overlapping cloud. Right: individual PCA on diagnostic markers — strong, structured differentiation.

The neutral background is consistent with a single panmictic pool — which sharpens the signal. Taking the low neutral F_{ST} to its limit: if the 20 populations are effectively *one* connected pool at neutral loci, then subsampling a single pool 20 times should reproduce the data — and it does. Among-population F_{ST} at neutral markers then sits at the small-sample floor (≈ 0.07 here; Hudson $\rightarrow \approx 0$), matching the empirical neutral value and explaining it more directly than 20 independently drifting demes (which over-differentiate to ≈ 0.08). But a single panmictic pool — like *any* neutral demography — is **DI-blind**: drift and sampling cannot tell ancestry-informative markers from uninformative ones, so they give the same F_{ST} at every DI. Figure 3 makes the point: the single-pool null and the 20-deme null are both flat across DI, while the data rise ~ 6 -fold. This is the strongest form of the argument — against a panmictic baseline the neutral differentiation is ≈ 0 , so there is essentially *no drift left to attribute the diagnostic F_{ST} to*: the observed 0.31 is almost entirely excess, and its DI-scaling can only come from a DI-specific process (ancestry sorting / selection). We therefore treat the hybrid populations as effectively panmictic at neutral loci (large N_e / connectivity) and read the diagnostic differentiation against that flat, near-zero baseline.

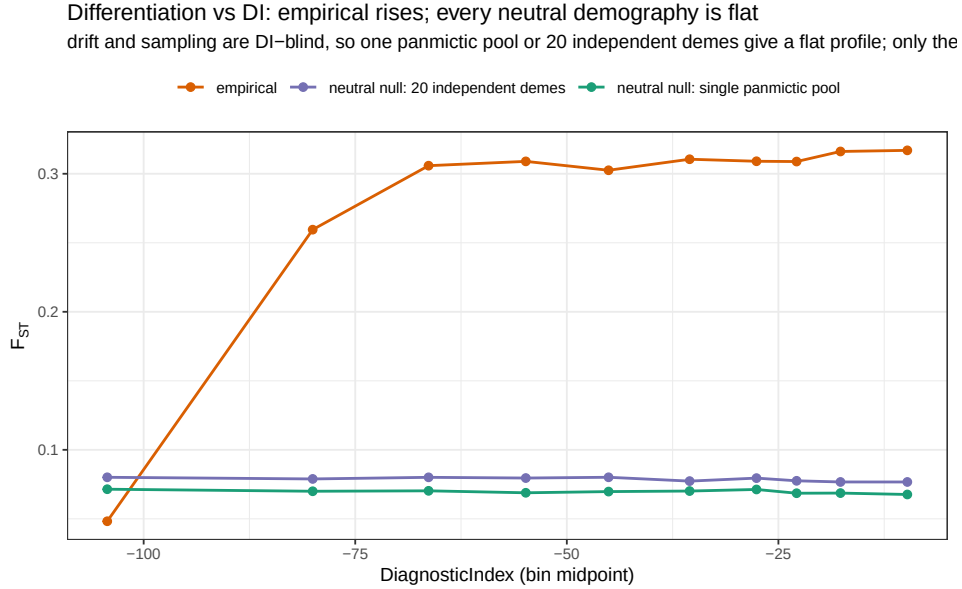


Figure 3: Among-population F_{ST} versus DiagnosticIndex. Every neutral demography is flat because drift and sampling are DI-blind: a single panmictic pool (≈ 0.07) and 20 independent demes (≈ 0.08) both give a horizontal profile. The empirical data rise ~ 6 -fold from the neutral floor to the diagnostic markers. No neutral model can be near-zero at neutral loci *and* strongly differentiated at diagnostic loci — the DI-scaling is the signal.

4 The diagnostic differentiation is locus-specific, not variable admixture

If the populations were simply founded at different aquilonia:polycтена ratios, each would carry a single ancestry proportion a_j and its diagnostic allele frequencies would follow $p_{ij} = f_i^{\text{pol}} + a_j \Delta p_i$ (rank-1). We fit this per population and decomposed the among-population variance at diagnostic markers into: sampling (binomial), the single ancestry axis, and a genuine locus-specific residual. Of the observed diagnostic $F_{ST} = 0.313$:

component	F_{ST} (fraction)
sampling noise	0.049 (16%)
single ancestry axis (variable admixture)	0.019 (6%)
locus-specific residual	0.245 (78%)

Per-population best-fit ancestry does vary (0.35–0.65, SD 0.077; Fig. 4 top), but that single knob explains only $\sim 6\%$ of the differentiation. Variable founding admixture is therefore rejected as the explanation.

Combining the two structure results: drift is excluded (neutral $F_{ST} \approx 0$) and variable admixture is excluded (6%), so the 0.245 locus-specific diagnostic differentiation is neither. This is where the internal control pays off: a neutral null calibrated to reproduce neutral $F_{ST} \approx 0$ has essentially no drift and therefore predicts ≈ 0 diagnostic differentiation, making the observed 0.245 *almost entirely excess over neutral*.

An honest wrinkle: divergent, not parallel. The among-population *variance* decomposed here measures *divergence* — populations differing at diagnostic loci. Parallel sorting (all populations fixing the same allele at a locus, Module A’s DI-governs-direction result) produces agreement, i.e. *low* among-population variance, and lives in the *mean* shift rather than this residual. So the 0.245 is a *divergent* locus-specific signal, and the residual PCA is correspondingly diffuse and high-dimensional (Fig. 4 bottom; PC1/PC2 $\approx 9\%$ each, though small per-population n confounds this view). Divergent ancestry differentiation without drift points toward local/divergent processes (the extrinsic/climate arm, Module C) more than a single shared intrinsic incompatibility axis. It does not contradict Module A — a parallel mean direction and divergent variance can coexist — but it says the *differentiation among populations* is dominated by locus-specific divergence.

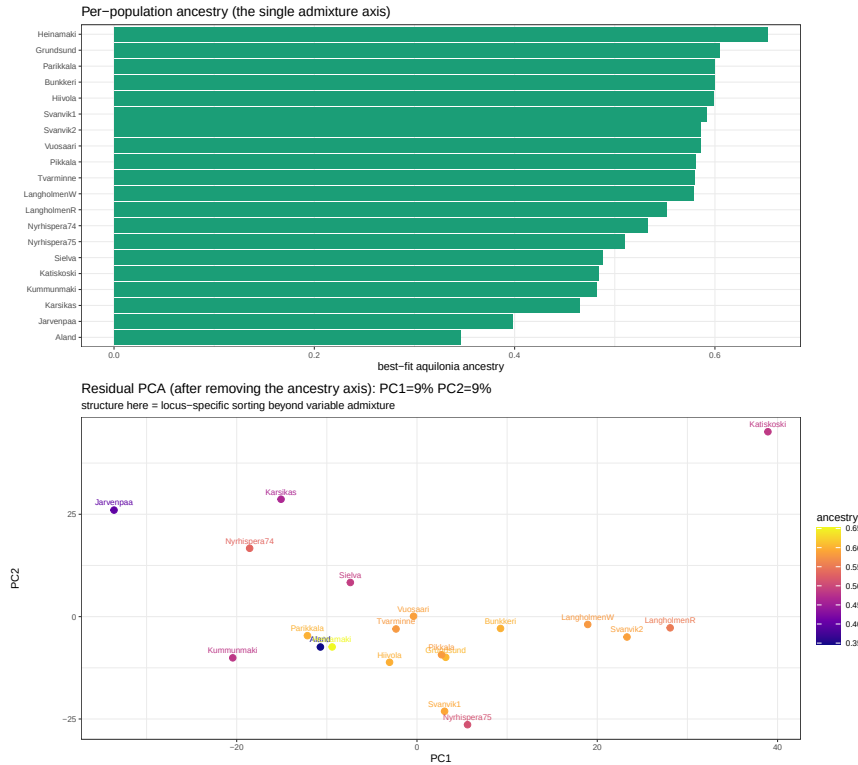


Figure 4: Top: per-population best-fit aquilonia ancestry (the single admixture axis), 0.35–0.65. Bottom: PCA of the residual after removing that axis; structure here would be locus-specific sorting beyond variable admixture. It is diffuse and high-dimensional (sampling-confounded at these sample sizes).

5 The residual null–data LD mismatch is a founder-diversity ceiling

The null reproduces the recombination-driven LD *landscape* well: local LD (ld_w , median r^2 in a 100 kb window) peaks on the same chromosomes as the data, per-chromosome Spearman 0.92–0.93, and this is robust across generations, foundings, and K (Fig. 5). What it does *not* reproduce is the *level*: the null’s local LD sits about $2\times$ above the data (mean $ld_w \sim 0.20$ vs 0.09; median 0.10 vs 0.02). We traced this excess to its source by elimination, and the answer is not a knob we can turn in the model.

Not the founding bottleneck. Seeding each deme from only ~ 43 real founders and growing to K is a severe bottleneck. We made it shallow by resampling founders *with replacement* from an inflated pool (each real haplotype written $40\times$), so the founder census — and hence the founding F_{ST} — became a free knob while genetic diversity stayed capped at the real 30/13 haplotypes. A shallower founding roughly halves the neutral F_{ST} ($0.15 \rightarrow 0.076$) and recovers most of the π deficit ($0.40 \rightarrow 0.437$), confirming the bottleneck was a genuine contributor to over-differentiation (Table 2). But the local-LD level does not shift *at all* across foundings, and a residual neutral F_{ST} (0.076 vs data 0.048) and π deficit persist even at the whole-pool limit.

Not drift / too-small N_e . We then swept K *downward* (781–6250) at fixed founding (450/195). Neutral F_{ST} scales cleanly with $1/K$ — confirming it is partly drift — but its $K \rightarrow \infty$ intercept is 0.073, still above the data’s 0.048; and the local-LD level is essentially *flat* across the entire K range at every quantile (median ld_w 0.103 $\rightarrow$ 0.100 over an $8\times$ change in K ; Fig. 6). Background (inter-chromosomal, unlinked) LD *does* scale with $1/K$ as drift theory predicts, but it is already *below* the empirical value. So neither the local-LD excess nor the residual F_{ST} is a large- N_e target: raising K would not close them. This supersedes an earlier reading of the LD floor as a Wahlund / too-small- N_e effect.

Not the recombination map. The map is high, not low — 15.7 cM/Mb genome-wide (4,696 cM over 300 Mb; 1 cM $\approx$ 64 kb), correctly scaled. And the null and data LD-decay curves *converge to the same background* ($r^2 \approx 0.033$) by ~ 1 Mb (Fig. 7); too-low recombination would leave the null elevated at long range, which it is not. The excess is confined to *short* range ($2\times$ at 65 kb, gone by 1 Mb), exactly where founder LD is immune to recombination: at 4 kb the recombination fraction is only ~ 0.06 cM, so $\sim 96\%$ of founder LD survives 60 generations regardless of the rate.

What it is. By elimination, the residual short-range LD excess (and the associated π deficit and residual F_{ST}) is the *finite founder-haplotype panel*: with only 30 *aquilonia* and 13 *polycтена* founding haplotypes, nearby markers carry more LD than a founding drawn from the large, many-haplotype real parental populations, and no amount of population size, founder census, or recombination can manufacture the missing haplotype diversity. It is liftable only by sequencing more parents or adding variation (mutation and burn-in) — the very thing the conditional design deliberately avoids. Crucially this is a *short-range LD* artefact: it leaves the DI dose-response (Sec. 2) and the sorting classification unchanged, and the null’s diagnostic-marker F_{ST} stays flat across every one of these manipulations, so the calibrated-null sorting test is unaffected by it.

6 Where this stands, and next steps (*WIP*)

Two demographic explanations for the low-neutral / high-diagnostic dissociation are now excluded from the empirical data alone: *drift* (neutral $F_{ST} \approx 0$, and drift is DI-blind) and *variable founding admixture* (6% of the diagnostic differentiation). What remains is a large ($F_{ST} \approx 0.245$), locus-specific, *divergent* ancestry signal against a near-zero neutral expectation. The simulation’s role

Table 2: Founder-number sweep at $K = 6250$ (15k DI-stratified markers, 20 demes, gen 60). Making the founding shallow (resampling to a larger census from the same 30/13 haplotypes) removes about half the neutral F_{ST} excess and most of the π deficit, then plateaus; the local-LD level (last column) does not respond.

founding (N_{aq}/N_{pol})	neutral F_{ST}	neutral π	mean ld_w
30/13 (deep, baseline)	0.15	0.40	0.197
150/65	0.091	0.429	0.196
450/195	0.079	0.436	0.197
1200/520 (whole pool)	0.076	0.437	0.196
empirical	0.048	0.450	0.089

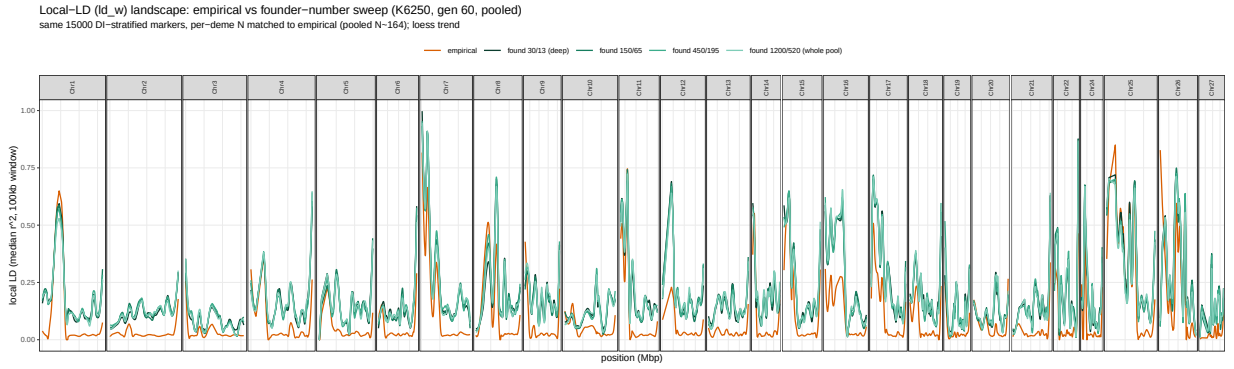


Figure 5: Local LD (ld_w) along the genome, empirical vs the founder-number sweep (K=6250, gen 60, pooled, per-deme N matched to the data). The recombination-driven landscape is reproduced (peaks co-localise, per-chr Spearman 0.92–0.93); the null curves for all foundings lie on top of one another well above the empirical level — the excess does not respond to the founding bottleneck.

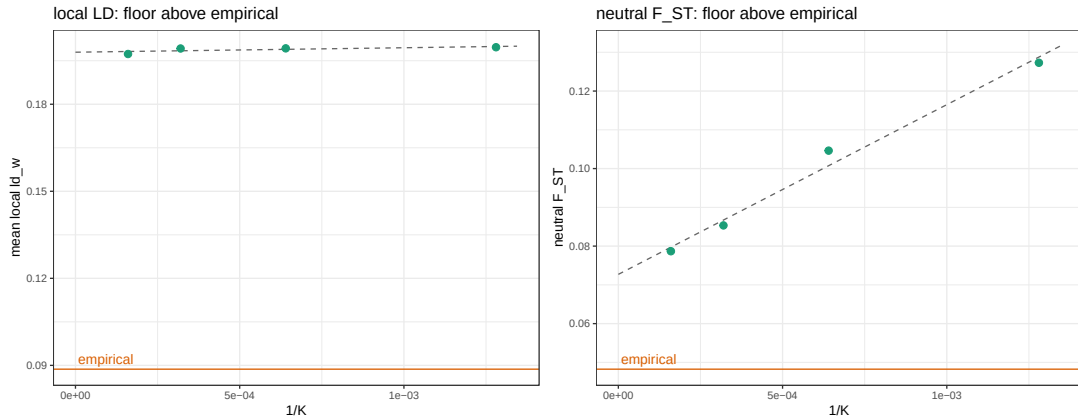


Figure 6: Sweeping K downward at fixed founding (450/195, gen 60), each statistic against $1/K$. Left: mean local ld_w is flat across an $8\times$ change in K (the $K \rightarrow \infty$ floor sits far above the empirical line) — the LD excess is not drift. Right: neutral F_{ST} is linear in $1/K$ (partly drift) but extrapolates to 0.073 at $K \rightarrow \infty$, still above the empirical 0.048.

LD decay vs physical distance: empirical vs null (450/195, K=6250, gen 60)
same 15k markers, within-chromosome pairs, MAF $\geq$ 0.1, matched N; curves converge by $\sim$ 1 Mb

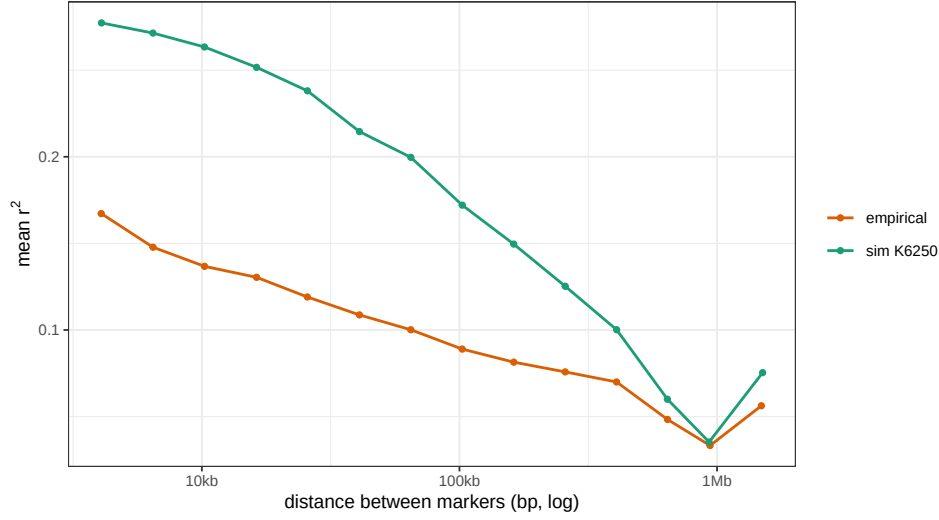


Figure 7: LD decay versus physical distance, empirical vs null (450/195, K=6250, gen 60; same markers, within-chromosome pairs, matched N). The two curves converge to the same background by $\sim$ 1 Mb — large-scale recombination matches the data — while the null carries excess LD only at short range, the signature of the finite founder-haplotype panel.

is now sharply defined: calibrate the null to reproduce neutral $F_{ST} \approx 0$ (the very-large- N_e / effectively-panmictic-at-neutral regime), at which point the diagnostic excess is read against ≈ 0 .

1. **Calibrated-null p -value.** Run the sorting statistic (`parallelism_stats`) through the null at the best-achievable calibration — shallow founding and K near the empirical N_e (Sec. 5). The null’s neutral differentiation is bounded below by the founder-diversity ceiling (sampling plus limited founder haplotypes, not drift), so the neutral floor is used as the reference rather than driven to zero; the diagnostic dose-response, which is invariant to founding, K , and recombination, carries the test. This puts a p -value on the excess and separates the parallel (mean) from the divergent (variance) components. *Lifting the ceiling itself (more sequenced parents, or adding mutation with a burn-in) is a separate, larger undertaking.*
2. **Divergent vs parallel.** Characterise the residual structure with individual-level data (controlling sampling), and test whether it aligns with climate (reconciling with Modules A and C: parallel direction vs divergent differentiation).
3. **Non-independence.** Neutral $F_{ST} \approx 0$ is degenerate between enormous N_e and connectivity/shared ancestry. A one-time *founding tree* is ruled out: drawing the 20 populations from only 15/10/5 independent demes leaves every statistic unchanged (co-descension replicates frequency vectors without changing their variance), and the empirical neutral markers show no clustering into founding groups. So the low neutral F_{ST} points to large N_e / ongoing connectivity (effective panmixia, Fig. 3) rather than a shared-founding hierarchy. What remains open is the *effective number of independent replicates* for the sorting p -value, to be estimated from the empirical neutral coancestry (cf. Nohaud et al.).

All analyses run from a 19.5 MB compact empirical bundle (replacing the 1 GB dataset); scripts and the portable simulation pipeline are in `moduleE.slim/` in the repository.